
GIL Clinic — Complete System

User Manual & Module Guide

43 Utility Modules + 61 Page Modules = 104 Total Modules

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1. System Overview

CardioQueue is a comprehensive hospital management system designed for GIL Clinic. It handles patient registration, department queue management, billing, pharmacy, inventory, HR, finance, AI-assisted tools, and multi-channel communication — all in one Streamlit web application.

Architecture

UI Pages → `llm_harness.py` → `db.py` (SQLite + Google Sheets + Local JSON)

- Frontend: Streamlit (multi-page, role-based navigation)
- Orchestration: **`llm_harness.py`** — central action router
- Database: SQLite primary, Google Sheets fallback, local JSON cache
- PWA: `manifest.json` + `service-worker.js` for Add-to-Home-Screen
- Notifications: Browser push, WhatsApp (Meta Cloud API), SMS, Email

How to Start

Run: **`streamlit run app.py`** from the project root. Open the URL shown in terminal (default `http://localhost:8501`).

2. Login & Access Control

Login Flow

- On first load, you see a gradient card grid of staff members grouped by role.
- Click your name → enter your PIN (4-6 digits).
- Admin: click name → enter password **gurjas@123**.
- Patients: click "Patient" button at bottom → goes to self-service status page.
- Admin toggle link at bottom switches to "Password Login" mode (for keyboard entry).

Role-Based Access (RBAC)

Every page is protected by role. The system has 12 roles with granular permissions. See *pages/RBAC.py* for the permission matrix viewer.

3. Role-Based Modules

3.1 Admin Dashboard

- **Page:** Admin / Admin_Panel
- Full system KPIs: total patients today, revenue, pending reports
- User management: add/edit staff, reset PINs
- System-wide settings and configuration
- Access all modules across all departments

3.2 Reception / Registration

- **Pages:** Registration, Reception, Appointments
- Register new patients: name, mobile, age, gender, address, doctor reference
- Book appointments (walk-in or scheduled)
- Assign tests (ECG, Echo, TMT, OPD, Lab, X-Ray, Ultrasound)
- Print token slip with department, test name, token number, wait time
- Bulk import patients from CSV via *pages/BulkImport.py*

3.3 Doctor / OPD

- **Pages:** Doctor, OPD, FollowUp
- View all patients assigned for consultation
- Add OPD consultation notes per test
- Prescribe medicines → flows to Pharmacy/Billing
- View patient history, past reports, timeline
- Mark consultation complete → patient moves to next stage

3.4 Department Staff (ECG, Echo, TMT)

- **Pages:** ECG, Echo, TMT
- Live queue of patients waiting for that test
- Call next patient (triggers WhatsApp notification + beep)
- Mark test in progress / complete
- Auto-refresh every 5 seconds
- Room assignments per technician

3.5 Nurse Station

- **Page:** Nurse
- View all today's patients across all departments
- Record vitals (BP, pulse, temperature, SpO2, weight)
- Filter patients by waiting/called/in-progress status

3.6 Pharmacist & Pharmacy

- **Pages:** Pharmacist, Pharmacy

- View pending prescriptions from doctors
- Dispense medicines (auto-deduct from inventory)
- Low stock alerts and expiring batch warnings
- Medicine master: add/edit medicines with batch tracking

3.7 Lab Technician & Radiology

- **Pages:** Lab_Technician, Lab, Radiology, XRay, Ultrasound
- Sample collection workflow: collect → process → verify → report
- Upload test results / reports (PDF, images)
- Mark report ready → notifies patient via WhatsApp/SMS

3.8 Billing & Accountant

- **Pages:** Billing, Accountant, Finance
- Create bills with test items, medicines, discounts
- Auto-GST calculation (HSN/SAC codes)
- Payment modes: Cash, Card, UPI, Insurance
- View today's billing summary, pending payments
- P&L; dashboard, expense tracking, monthly summaries

3.9 HR & Payroll

- **Pages:** HR, Payroll
- Staff directory: all employees with roles, contact, salary
- Attendance tracking (present/absent/leave)
- Leave request management (apply/approve/reject)
- Monthly payroll: calculate salary, deductions, net pay
- Payslip generation

3.10 Manager / Owner Dashboard

- **Pages:** Manager_Dashboard, Owner_Dashboard, Reports, Analytics
- Daily, weekly, monthly revenue charts
- Patient volume trends, department-wise load
- Top tests performed, doctor referral analysis
- Export reports as CSV/PDF
- Owner dashboard: full business KPIs at a glance

4. Patient Modules

4.1 Patient Status (Self-Service PWA)

- **Page:** Patient_Status
- Mobile-first PWA: Add to Home Screen on phone
- Auto-load via QR scan: *?patient=ID*
- Or enter mobile number: *?mobile=NUMBER*
- Live status bar showing journey progress
- Sound + vibration when called or report ready
- Auto-refresh every 5 seconds
- Missed call alert detection
- Bilingual: Hindi + English

4.2 Patient Portal

- **Page:** Patient_Portal
- Self-service appointment booking (date, time, test type)
- View past appointments and test history
- Download reports
- Submit feedback

4.3 Patient Timeline & Tracking

- **Pages:** Patient_Timeline, Patient_Tracking
- Timeline: full journey from registration to report delivery
- Tracking: live flow monitor across all departments
- Search by Patient ID or mobile number

5. Communication Modules

5.1 Email

- **Page:** Email | **Utils:** utils/email.py
- Configure SMTP settings in the page
- Send single or bulk emails
- Report email templates (test results in HTML)
- Email log viewer with sent/failed status

5.2 SMS Manager

- **Page:** SMS_Upgrade | **Utils:** utils/sms_upgrade.py
- Create and manage SMS templates (DLT-compliant)
- Send appointment reminders, report alerts, promotional SMS
- Template variables: {patient_name}, {doctor}, {time}, etc.

5.3 WhatsApp Business

- **Page:** WhatsAppUpgrade | **Utils:** utils/whatsapp_upgrade.py
- Meta Cloud API integration (send template messages)
- Send appointment confirmations, call alerts, report ready notifications
- Configure PHONE_NUMBER_ID and ACCESS_TOKEN
- Legacy WhatsApp also available via utils/whatsapp.py

5.4 Push Notifications

- **Page:** PushNotifications | **Utils:** utils/push_notifications.py
- Register devices for browser push
- Send targeted or broadcast notifications
- Notification categories: appointment, report, promotion, alert

5.5 Voice Call & Video Call

- **Pages:** VoiceCall, VideoCall
- Voice: initiate calls, view call history, call logs
- Video: create telemedicine rooms, start/end sessions
- Room management for doctor-patient video consultations

6. AI Modules

6.1 AI Dietician

- **Page:** AI_Dietician | **Utils:** utils/ai_dietician.py
- Generate personalized meal plans based on:
 - - Disease/condition (diabetes, hypertension, heart disease, etc.)
 - - Dietary preference (vegetarian, vegan, non-veg)
 - - Calorie target
- Pre-defined diet plans for common conditions
- Custom meal plan generation

6.2 AI Report Explainer

- **Page:** AI_Report_Explainer | **Utils:** utils/ai_report_explainer.py
- Enter test name + value → get plain-English explanation
- Shows reference ranges, flag (normal/high/low)
- Contextual advice based on deviation
- Covers 50+ medical tests with REFERENCE_RANGES

6.3 AI Prescription

- **Page:** AI_Prescription | **Utils:** utils/ai_prescription.py
- Medicine database with dosages, contraindications
- Drug-drug interaction checker
- Suggest medicines for common conditions
- Generate formatted prescription with doctor info

6.4 AI Voice Agent

- **Page:** AI_VoiceAgent | **Utils:** utils/ai_voice_agent.py
- Voice agent session management
- Agent types: appointment booking, general inquiry, prescription refill
- Create and monitor voice agent sessions

6.5 AI Triage & AI Receptionist

- **Pages:** AI_Triage, AI_Receptionist
- AI Triage: symptom assessment → recommend department priority
- AI Receptionist: automated patient check-in and guidance

7. System & Admin Tools

7.1 Backup & Restore

- **Page:** Backup | **Utils:** utils/backup.py
- Create database backups on demand
- View backup history with timestamps and sizes
- Backup stats: total size, count, last backup

7.2 Encryption & Audit

- **Page:** EncryptionPage | **Utils:** utils/encryption.py, utils/audit.py
- Encryption: AES-256-GCM via Fernet, encrypt/decrypt text, PII protection
- Audit: SHA-256 chained audit log, verify chain integrity
- Every action logged immutably

7.3 RBAC & Compliance

- **Pages:** RBAC, Compliance
- RBAC: 12-role permission matrix viewer, check permissions
- Compliance: patient consent management, data rights requests
- Consent types: treatment, data sharing, research, billing

7.4 Monitoring & Logging

- **Pages:** Monitoring, Logging
- Monitoring: system health dashboard, DB size, page count, metrics
- Logging: centralized log viewer with search, filter by level
- Record and review all system events

7.5 Multi-Branch

- **Page:** MultiBranch | **Utils:** utils/multi_branch.py
- Add and manage multiple clinic branches
- View all branches in the network

8. Finance & Inventory

8.1 Finance (P&L;)

- **Page:** Finance, Accountant, Owner_Dashboard | **Utils:** utils/finance.py
- Profit & Loss dashboard with revenue vs expenses
- Add expenses (rent, salary, utilities, supplies)
- Monthly summary with category breakdown
- Expense categories: 10+ predefined

8.2 GST Compliance

- **Page:** GST | **Utils:** utils/gst.py
- HSN/SAC code maps for medical services and products
- Auto-GST calculation (5%, 12%, 18%, 28%)
- GST invoice generation
- Tax summary reports

8.3 Inventory & Purchase

- **Page:** Inventory, Purchase | **Utils:** utils/inventory.py, utils/purchase.py
- Inventory categories, items, batch tracking (FEFO)
- Stock movements: in/out/transfer/adjustment
- Low stock alerts, expiring batch warnings
- Purchase orders with approval workflow
- Stock audit: create audit session → count → resolve variances

8.4 Vendor Management

- **Page:** Vendor | **Utils:** utils/vendor.py
- Vendor registry with GST, contact, ratings
- Track purchases by vendor

9. IPD & Emergency

IPD (In-Patient Department)

- **Page:** IPD_Ward | **Utils:** utils/ipd.py
- Admit patients to wards/beds
- Bed management: view occupied/available beds
- Discharge summary generation
- Track IPD billing (room rent, procedures, medicines)

Emergency

- **Page:** Emergency | **Utils:** utils/emergency.py
- Rapid patient registration in emergency mode
- Priority-based queue (critical → serious → stable)
- Quick vitals recording

10. Quick Reference — Default Credentials

Role	Name	PIN / Password
Admin	Admin	gurjas@123
Reception	Receptionist	1234
ECG	ECG Technician	1234
Echo	Echo Technician	1234
TMT	TMT Technician	1234
OPD	OPD Staff	1234
Manager	Manager	1234

- PIN is 4-6 digit numeric (new users). Legacy text passwords also supported.
- Auto-seed: 7 default users created when database is empty.
- Change passwords/PINs from Admin panel.

Appendix A: Complete Module List

43 Utility Modules + 61 Page Modules = 104 Modules

Utility Modules — 42 total

ai_dietician
ai_followup
ai_prescription
ai_receptionist
ai_report_explainer
ai_triage
ai_voice_agent
appointments
audit
backup
billing
compliance
config
db

email
emergency
encryption
feedback
finance
followup
gst
hr
inventory
ipd
lab
local_json_db
logging
monitoring

multi_branch
notifications
payroll
purchase
push_notifications
queue
rbac
sms
sms_upgrade
vendor
video_call
voice_call
whatsapp
whatsapp_upgrade

Page Modules — 61 total

AI_Dietician
AI_FollowUp
AI_Prescription
AI_Receptionist
AI_Report_Explainer
AI_Triage
AI_VoiceAgent
Accountant
Activity_Log
Admin
Analytics
Appointments
Backup
Billing
Compliance
Daily_List
Doctor
ECG
Echo
Email
Emergency

EncryptionPage
Feedback
Finance
FollowUp
GST
HR
IPD_Ward
Inventory
Lab
Lab_Technician
Logging
Manager
Monitoring
MultiBranch
Nurse
OPD
Owner_Dashboard
Password_Management
Patient_History
Patient_Portal
Patient_Status

Patient_Timeline
Patient_Tracking
Payroll
Pharmacist
Pharmacy
Purchase
PushNotifications
RBAC
Reception
SMS_Upgrade
TMT
Ultrasound
Vendor
VideoCall
VoiceCall
WhatsAppUpgrade
XRay
__init__
_department_base